Interpretation of Liver Function Tests (LFT)

A comprehensive guide to hepatocellular enzymes, synthetic function markers, heme metabolism, and clinical jaundice differentiation



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Learning Objectives:

Heme Metabolism

Describe heme degradation pathways and identify normal serum levels.

Bilirubin Fractionation

- Differentiate conjugated vs unconjugated serum bilirubin fractions
- Identify normal serum bilirubin level

Jaundice Classification

Correlate serum bilirubin pattern with pre-hepatic, hepatic, and post-hepatic jaundice.

Enzyme Pathology

Interpret specific liver enzymes (ALT, AST, ALP, GGT) in diagnosis.



Describe the degradation of Heme

- **Heme** is the iron-containing part of hemoglobin in red blood cells (RBCs). When old RBCs are destroyed,their hemoglobin is broken down & heme is converted into bilirubin.

Old red blood cells

↓

- Destroyed mainly by macrophages in the spleen, liver, bone marrow

↓

- Hemoglobin

↓

- Separated into globin and heme

↓

- Heme → Biliverdin → Unconjugated bilirubin

↓

- Liver conjugates bilirubin

↓

- Makes bilirubin water-soluble

↓

- Bile → Intestine → Feces

- Most bilirubin products leave the body through stool

- Heme oxygenase converts heme into biliverdin.

- Biliverdin reductase converts biliverdin into unconjugated bilirubin.

- Unconjugated bilirubin travels in the blood attached to albumin.

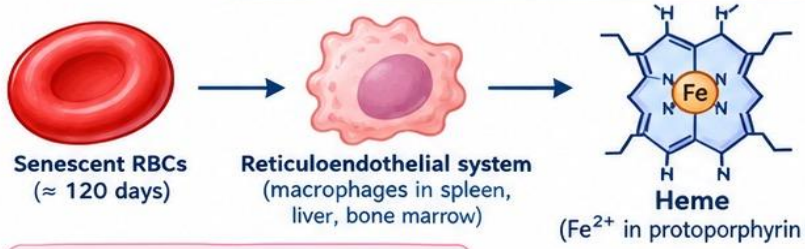
- In the liver, the enzyme UDP-glucuronosyltransferase (UGT1A1) conjugates bilirubin with glucuronic acid.

- Conjugated bilirubin is excreted into bile and eventually reaches the intestine.

- **Clinical importance:** If bilirubin production increases or the liver cannot process or excrete it properly, bilirubin can accumulate in the blood and cause jaundice.

Heme Degradation

Heme from old RBCs is broken down into biliverdin, bilirubin and finally excreted as urobilinogen and stercobilin.



Senescent RBCs (≈ 120 days) → **Reticuloendothelial system** (macrophages in spleen, liver, bone marrow) → **Heme** (Fe^{2+} in protoporphyrin IX)

Key Products

- **Biliverdin** (green pigment)
- **Bilirubin** (yellow pigment)
- **Iron** (recycled)
- **Urobilinogen** (in urine & feces)
- **Stercobilin** (in feces)

Important Enzymes

- **Heme oxygenase** (rate-limiting step)
- Biliverdin reductase
- UDP-glucuronyl transferase

Sites

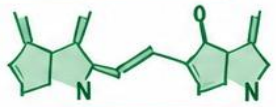
- Macrophages (heme breakdown)
- Liver (conjugation)
- Intestine (bacterial conversion)
- Kidney (urobilin excretion)

Final Excretion

- **Urine:** urobilin (yellow color)
- **Feces:** stercobilin (brown color)

1 Heme oxygenase

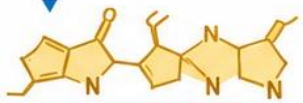
Heme is degraded by heme oxygenase (rate-limiting enzyme) using O_2 and NADPH.



Biliverdin (green pigment) + **Iron** (Fe^{2+}) (recycled and stored as ferritin or transferrin)

2 Biliverdin reductase

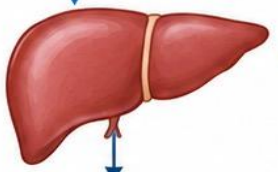
Uses NADPH to reduce biliverdin to bilirubin.



Bilirubin (yellow pigment, not water soluble)

3 Transport in blood

Bilirubin is transported to the liver bound to albumin.



Liver

4 Conjugation

In the liver, bilirubin is conjugated with glucuronic acid (by UDP-glucuronyl transferase) → water soluble.

Conjugated bilirubin (diglucuronide) → **Excreted in bile**

5 In intestine

Bacterial enzymes convert conjugated bilirubin to urobilinogen.

Urobilinogen (colorless)

- Majority → **Stercobilin** (in feces)
- **Reabsorbed** (small amount) → **Kidney** → **Urobilin** (in urine)

Remember:

- ✓ **Heme** degradation is a major source of iron recycling.
- ✓ **Bilirubin** is not water soluble until conjugated.
- ✓ **Urobilin** gives urine its yellow color; stercobilin gives feces its brown color.

Introduction to Liver Function Tests (LFTs)

- **What are LFTs?**

- Liver Function Tests are a group of blood tests that measure liver enzymes, bilirubin, and proteins to help evaluate the health of the liver.

- The liver performs several important functions:
- Metabolism: Processes carbohydrates, fats, & proteins.
- Detoxification: Helps process drugs & harmful substances.
- Bile production: Produces bile, which helps digest & absorb fats.
- Protein synthesis: Produces albumin & many clotting factors.
- Bilirubin processing: Converts unconjugated bilirubin into conjugated bilirubin for excretion.
- LFTs are blood tests used to assess liver cell injury, bile flow, and the liver's synthetic function.

- **Why are LFTs performed?**

1. Liver infections, such as hepatitis B and C

These infections can inflame and damage liver cells. ALT and AST may increase when damaged cells release these enzymes into the blood stream, indicating liver injury

2. Monitoring medication side effects

Certain medicines can damage liver cells or interfere with bile flow. LFTs help clinicians detect possible liver injury during treatment.

3. Evaluating other medical conditions

Diabetes, high triglycerides, and other metabolic conditions are associated with fatty liver disease. LFTs may help evaluate liver involvement.

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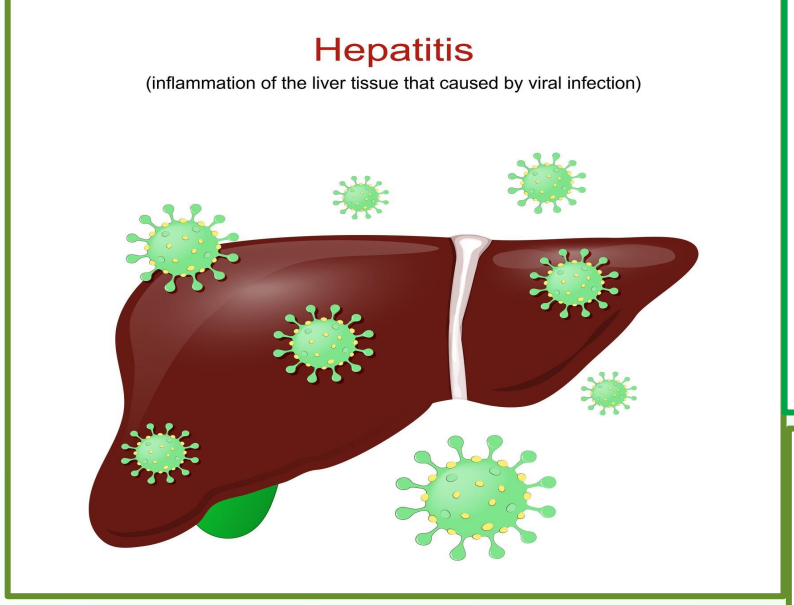
Anemia may also be investigated when increased RBC destruction could raise bilirubin.

4. Heavy alcohol consumption

Long-term excessive alcohol use can cause fatty liver, inflammation, and cirrhosis. AST, ALT, GGT, bilirubin, and albumin may help assess the condition.

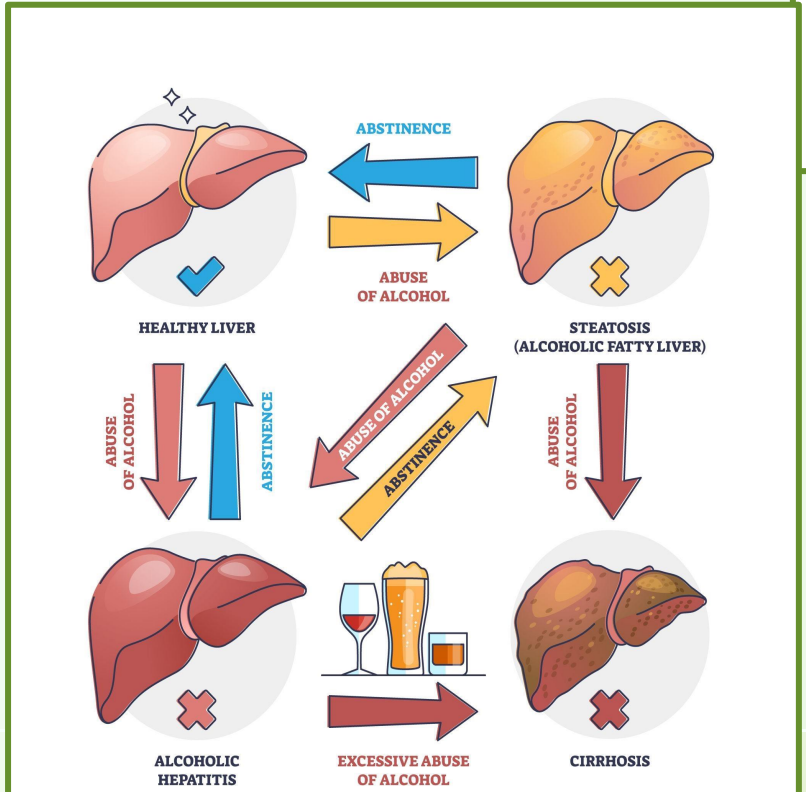
5. Gallbladder disease

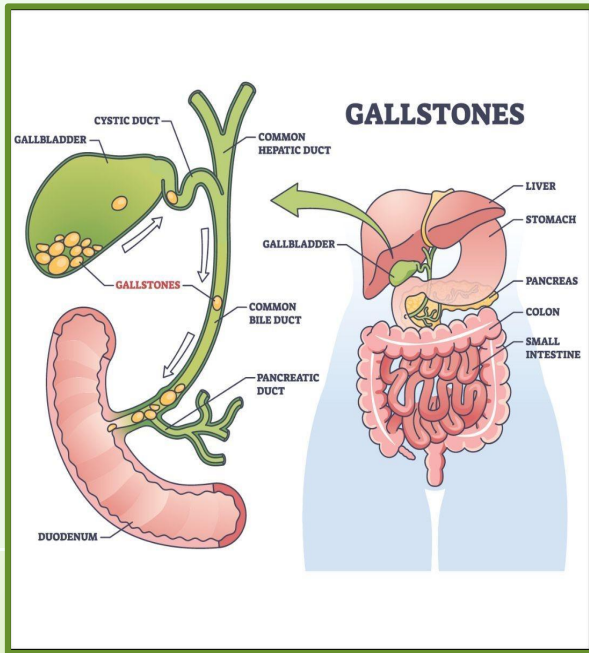
Gallstones may block the bile ducts and prevent normal bile flow. This can increase ALP, GGT, and conjugated bilirubin.





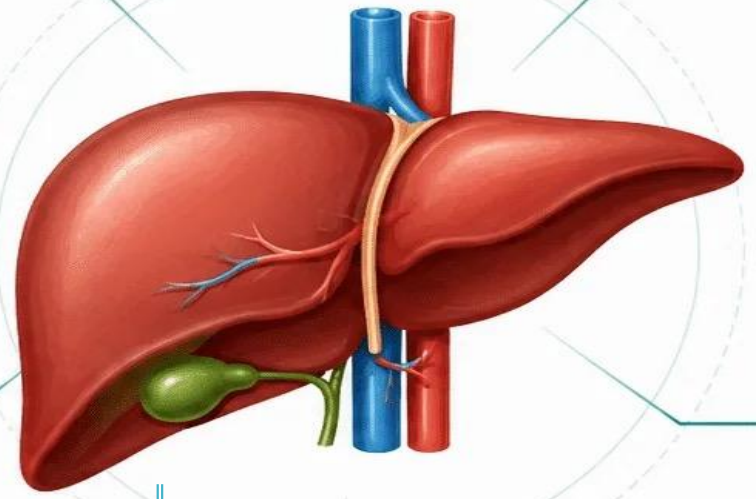






LIVER FUNCTION TESTS

Key Blood Markers for Liver Health



ALT

Alanine Aminotransferase

Indicates liver cell injury or inflammation

AST

Aspartate Aminotransferase

Indicates liver cell injury or inflammation

ALP

Alkaline Phosphatase

Indicates bile duct obstruction or damage

Bilirubin

Bilirubin

Indicates liver's ability to process bilirubin

Total Protein & Albumin

Measures total protein and albumin levels. Indicates liver's synthetic function and nutritional status.

GGT

Gamma-Glutamyl Transferase

Indicates bile duct problems or alcohol-related liver stress

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- Commonly Used Liver Function Tests (LFTs)

Test	Main function / Location	Clinical significance of increased or decreased levels
ALT (SGPT)	Enzyme mainly found in liver cells; involved in amino acid metabolism.	Increased ALT indicates liver cell injury.
AST (SGOT)	Enzyme found in liver, heart, and muscles.	Increased AST may indicate liver or other tissue injury.
ALP	Enzyme found mainly in bile ducts and bones.	Increased ALP may indicate bile duct obstruction or cholestasis.
GGT	Enzyme found in liver and bile ducts, among other tissues.	Increased GGT may indicate biliary disease and help identify a liver source of raised ALP.
Albumin	Major protein synthesized by the liver.	Decreased albumin may occur in chronic liver disease, malnutrition, kidney disease, or inflammation.
Bilirubin	Yellow pigment formed during red blood cell heme breakdown.	Increased bilirubin may indicate hemolysis, impaired liver processing, or bile flow obstruction

- Commonly used liver function test

- 1. Alanine transaminase (ALT/SGPT)

An enzyme involved in amino acid metabolism. It is found mainly in liver cells. Increased blood ALT commonly indicates liver cell injury.

- 2. Aspartate transaminase (AST/SGOT)

An enzyme involved in amino acid metabolism. It is found in the liver, heart, & muscles. Increased AST may indicate liver or other tissue injury.

- 3. Alkaline phosphatase (ALP)

An enzyme found in the bile ducts and bones, among other tissues. Increased ALP may indicate bile duct obstruction or another cholestatic disorder.

- 4. Gamma-glutamyl transferase (GGT)

An enzyme present in the liver and bile ducts, as well as other tissues. It often rises in biliary disease and can help determine whether elevated ALP is liver-related.

- 5. Albumin

A major protein synthesized by the liver. A low level may occur in chronic liver disease, but it can also result from malnutrition, kidney disease, or inflammation.

- 6. Bilirubin

A yellow pigment formed mainly during heme breakdown. Increased bilirubin may indicate excessive RBC destruction, impaired liver processing, or obstruction of bile flow.

Alanine Transaminase (ALT/SGPT)

Alanine transaminase (ALT) is an enzyme that helps the body metabolize amino acids. It is present mainly inside liver cells.ALT is also called SGPT, which stands for **serum glutamic-pyruvic transaminase.**

- **Principle**

When liver cells are damaged, ALT can leak from the cells into the bloodstream. A blood test measures the amount of ALT present.

Common causes ALT increase include:

1. Viral hepatitis.
2. Fatty liver disease.
3. Drug-induced liver injury.
4. Alcohol-related liver disease.
5. Other conditions causing liver inflammation or cell injury.

- **Clinical Interpretation of ALT**

ALT Result	Possible Interpretation
Within the laboratory reference range	No ALT elevation detected; does not exclude all liver disease.
Mildly elevated	May occur in fatty liver disease, medication effects, or other conditions.
Markedly elevated	Can occur in acute hepatitis or significant liver cell injury.

- The reference range varies by laboratory. ALT must be interpreted alongside symptoms, other blood tests, medicines, and clinical history.
- ALT is generally more specific for liver cell injury than AST, but ALT alone cannot determine the severity of liver disease or the liver's overall functional

Alkaline Phosphatase (ALP)

- Alkaline phosphatase (ALP) is an enzyme that removes phosphate groups from certain molecules. It works best in an alkaline environment.
- It is found mainly in the liver and bile ducts, bones, and several other tissues.
- **Functions of ALP**

- In the liver: It is associated with the cells lining the bile ducts and helps with processes involving phosphate-containing compounds.

- In bones: It is produced by osteoblasts, the cells responsible for bone formation & mineralization.

- In the laboratory: Measuring ALP in blood helps investigate bile duct problems & certain bone disorders.

- **Why is the ALP test performed?**

An ALP test is usually performed alongside other blood tests to investigate liver, biliary, or bone disease.

- 1. Liver disease

ALP may increase in conditions affecting the liver and bile ducts, particularly when bile flow is impaired.

- 2. Bile duct obstruction

Gallstones, strictures, or tumors may block bile flow. ALP can rise because of the resulting cholestasis (reduced or blocked bile flow).

- 3. Bone disease

ALP can increase during active bone formation and in certain bone disorders, such as Paget disease



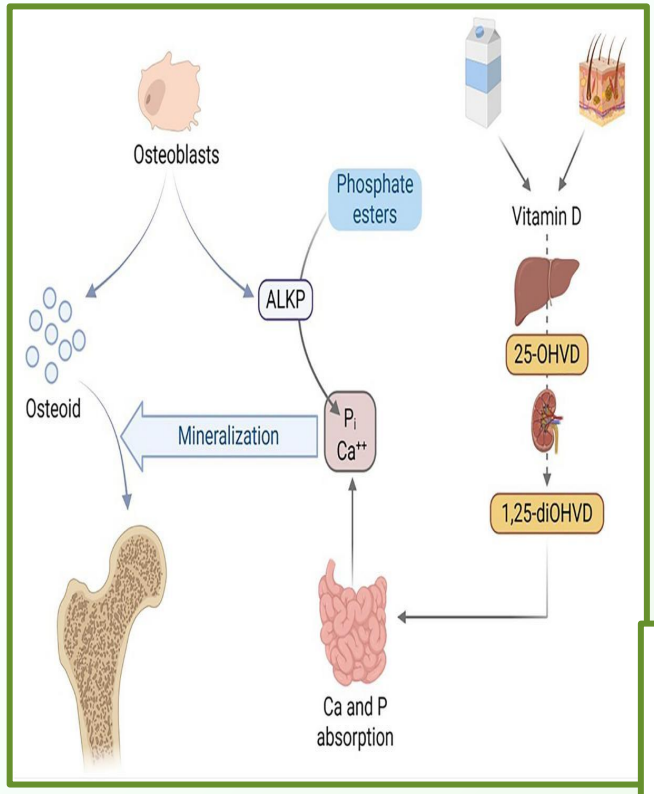
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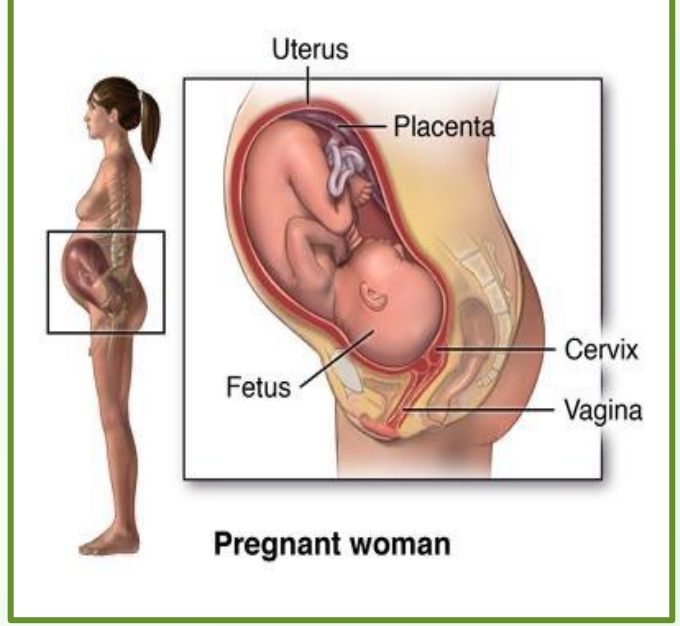
- 4. Children and adolescents

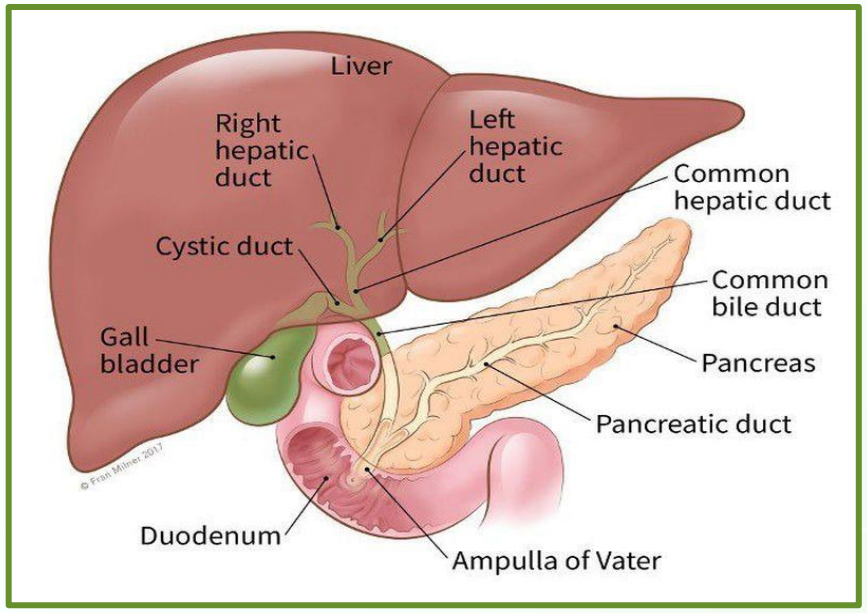
Growing bones have high osteoblast activity. Therefore, ALP may be naturally higher in children and teenagers than in adults.

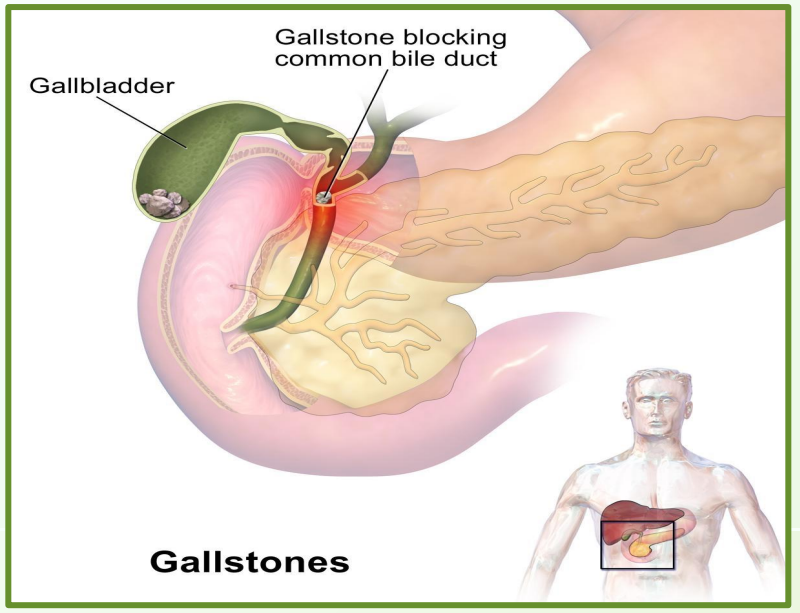
- 5. Pregnancy

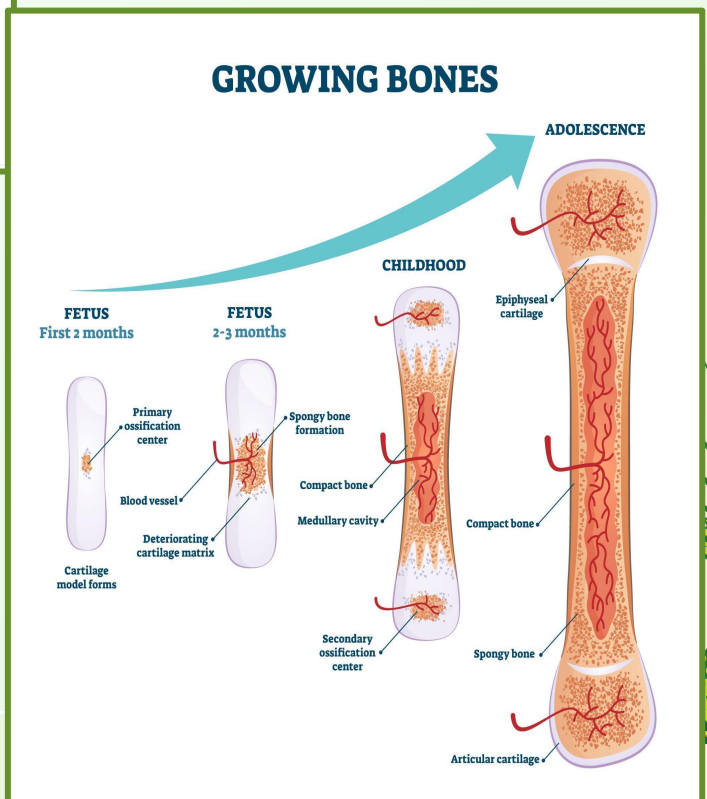
ALP may rise because the placenta produces its own form of ALP, particularly during later pregnancy.





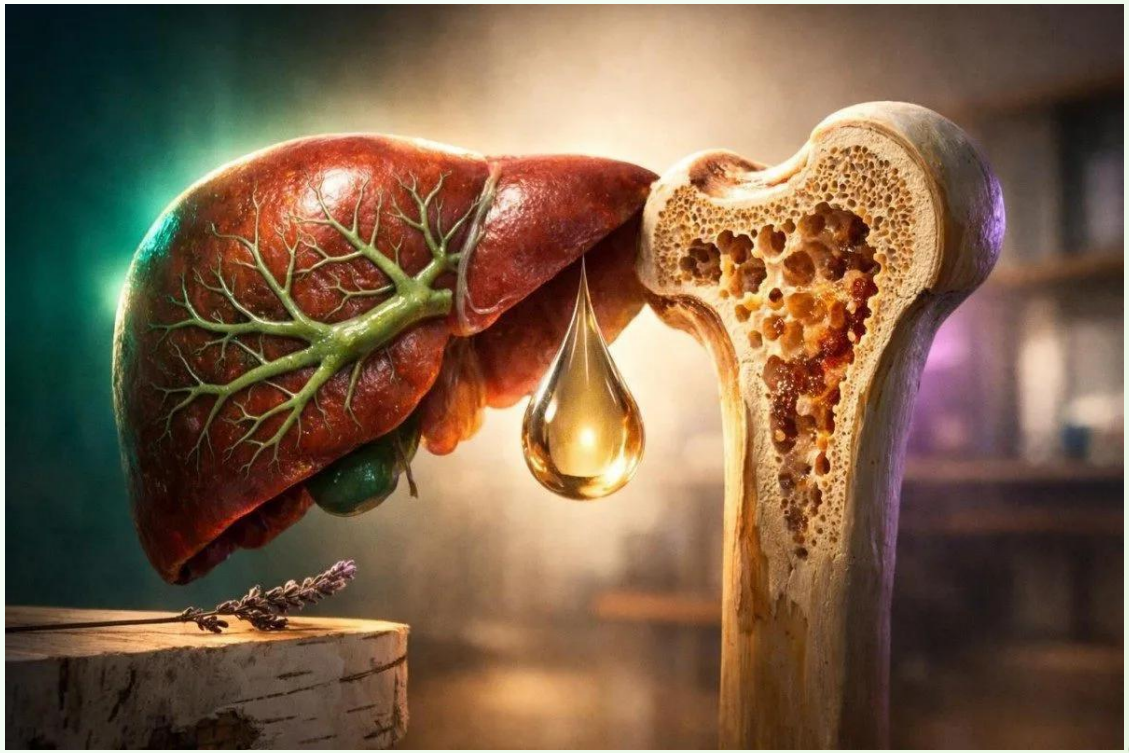






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Clinical interpretation of ALP



- Increased ALP + increased GGT: Suggests a liver or biliary source, especially cholestasis.
- Increased ALP + normal GGT: May suggest a bone source, although additional tests may be needed.
- Increased ALP in a growing child: May be a normal physiological finding.

ALP is not specific to the liver because it is also produced by bone and other tissues.

Gamma-Glutamyl Transferase (GGT)

Gamma-glutamyl transferase (GGT) is an enzyme involved in the metabolism and transfer of gamma-glutamyl groups, particularly in glutathione metabolism.

- It is present in several organs, with important concentrations in the liver and bile ducts.
- **Function of GGT**
- GGT helps break down and recycle glutathione-related compounds. Glutathione is an important antioxidant that protects cells against oxidative damage.
- In laboratory medicine, GGT is particularly useful for investigating liver and biliary disorders.

- **Why does GGT increase?**
- GGT can rise when liver cells are injured or bile ducts become obstructed.

- Normal liver and bile ducts

(GGT is present at relatively low blood levels)

↓

- Liver injury or bile duct obstruction

(GGT production or release into blood may increase)

↓

- Increased blood GGT

(Detected by a biochemical laboratory test)

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- **Causes of increased GGT**
- 1. Bile duct obstruction: Gallstones or tumors can interfere with bile flow.
- 2. Hepatitis: Inflammation may increase GGT.
- 3. Alcohol consumption: GGT may rise with heavy alcohol use, but it cannot diagnose alcohol-related disease by itself.
- 4. Medication effects: Certain medicines can increase GGT.
- 5. Other liver diseases: Fatty liver and other hepatobiliary disorders may elevate the enzyme

- **Why is GGT measured with ALP?**
- GGT helps determine whether an elevated ALP is likely to come from the liver or bile ducts rather than bone.
-

ALP Result	GGT Result	Possible Interpretation
High	High	Supports a liver or biliary (bile duct) source.
High	Normal	Consider a bone source or another non-hepatic cause.
Normal	High	May occur in several liver conditions, alcohol exposure, or medication effects

- **Important:** GGT is sensitive to many liver and biliary conditions, but it is not highly specific. An elevated GGT alone does not prove bile duct obstruction.

Aspartate Aminotransferase (AST/SGOT)

- Aspartate aminotransferase (AST) is an enzyme involved in amino acid metabolism. It is present in the liver, heart, skeletal muscles, and other tissues
- **Functions and clinical importance**
- AST transfers an amino group from aspartate to another molecule during amino acid metabolism.
- When cells are damaged, AST leaks into the bloodstream, causing its level to increase.
- **Causes of increased AST**
- - Liver injury: Hepatitis, fatty liver disease, & other liver disorders.
- - Muscle injury: Muscle trauma, strenuous exercise, or muscle diseases.

- Heart injury: Damage to heart muscle can also increase AST

Why is AST measured with ALT?

- ALT (alanine aminotransferase) is more specific to the liver than AST.Both enzymes are measured together because their levels help doctors assess possible liver cell

Feature	AST	ALT
Full name	Aspartate aminotransferase	Alanine aminotransferase
Main locations	Liver, heart, muscles	Mainly liver
Specificity for liver injury	Less specific	More specific
Increased level	Liver or other tissue injury	More suggestive of liver cell injury

Albumin

- Albumin is the main protein produced by the liver and is the most abundant protein in blood plasma.

PRIMARY HEPATIC PROTEIN

Albumin
Direct measure of liver synthetic capacity & long-term metabolic health.

Oncotic Pressure Maintenance

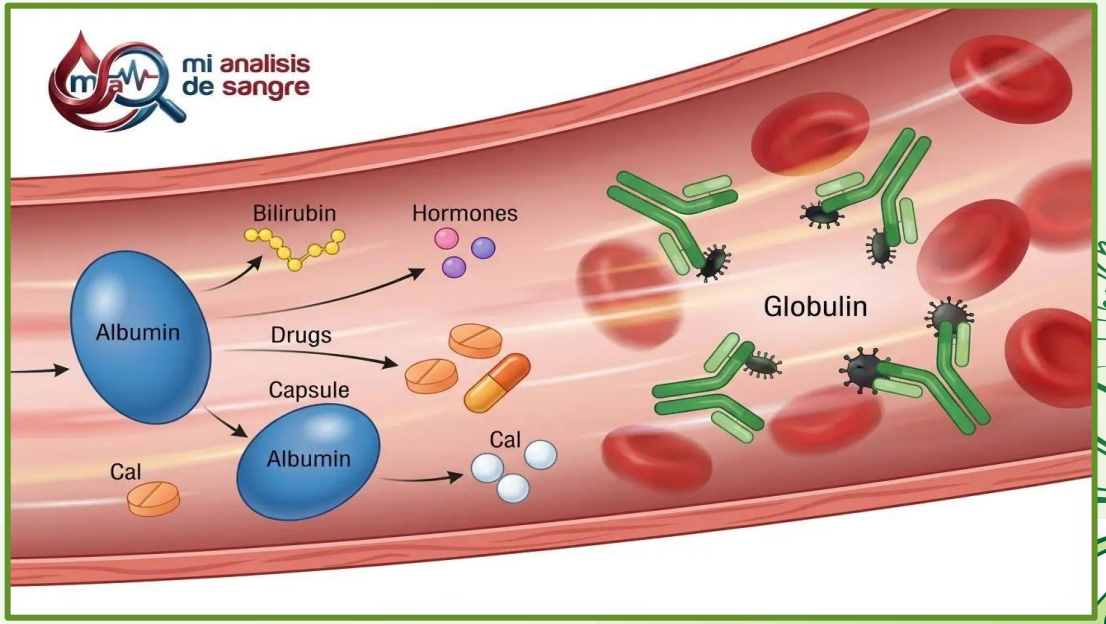
Prevents fluid leakage from intravascular blood vessels into surrounding tissues.

Molecular Transport Vehicle

Binds and carries hydrophobic hormones, vitamins, unconjugated bilirubin, and drugs.

Non-Hepatic Causes of Hypoalbuminemia

Malnutrition, nephrotic syndrome, systemic infection, and active inflammation

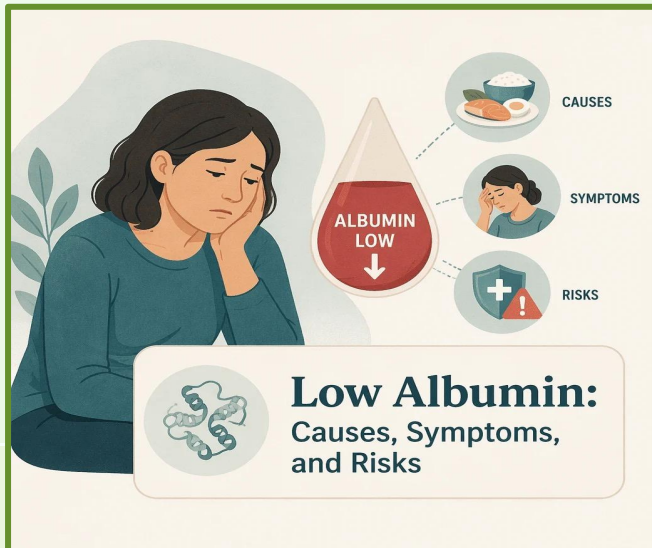


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- **Causes of decreased albumin**
- - Chronic liver disease: The liver may produce less albumin.
- - Kidney disease: Albumin may be lost in urine.
- - Poor nutrition: Inadequate protein and energy intake may contribute.
- - Inflammation or infection: Albumin can decrease during inflammatory conditions.
- - Protein loss from the digestive tract: Certain intestinal disorders can cause albumin loss.
- **Key point:** AST and ALT mainly help detect liver cell injury, whereas albumin helps assess the liver's ability to synthesize proteins

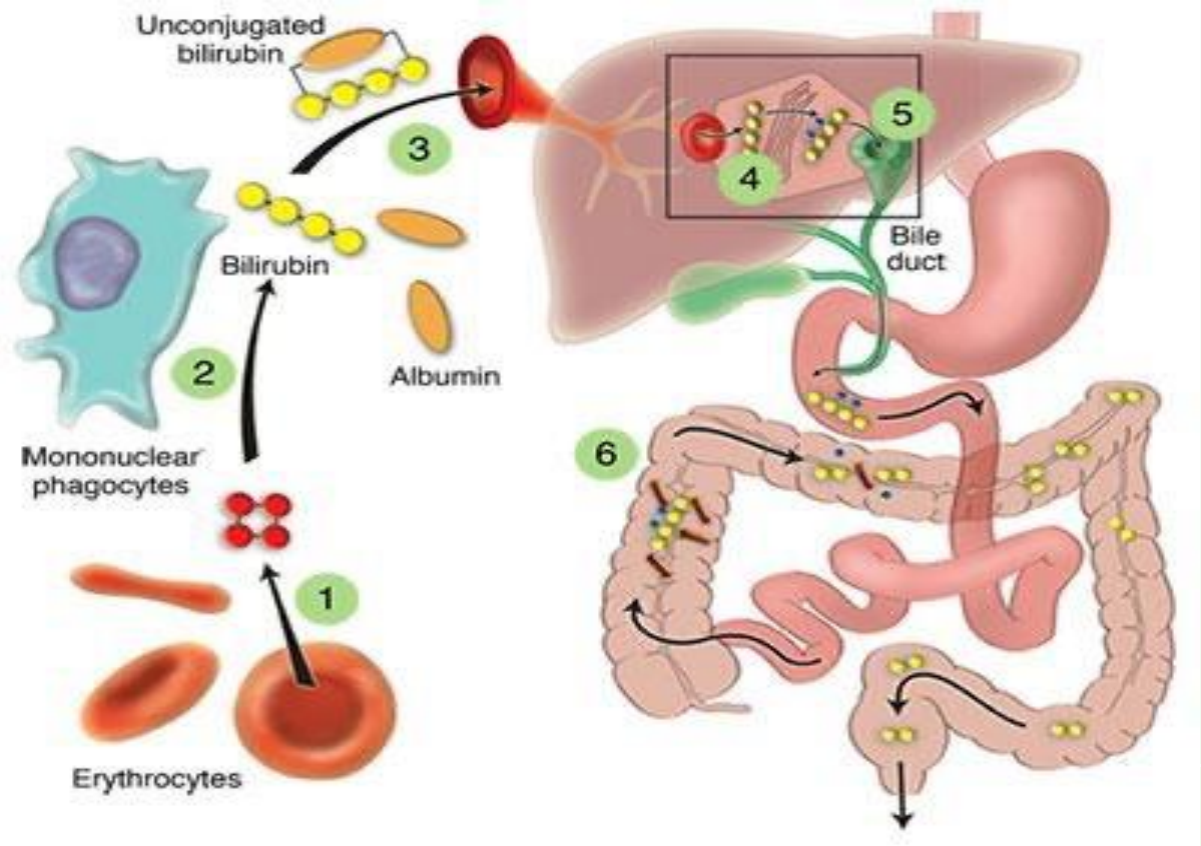
- **Clinical significance**
- Low albumin may be associated with swelling, fluid accumulation, or ascites, particularly when the reduction is substantial.
- Albumin is useful for assessing the liver's synthetic function, especially in chronic liver disease. However, a low level does not automatically mean that the liver is diseased because kidney disease, inflammation, and nutritional problems can also lower it

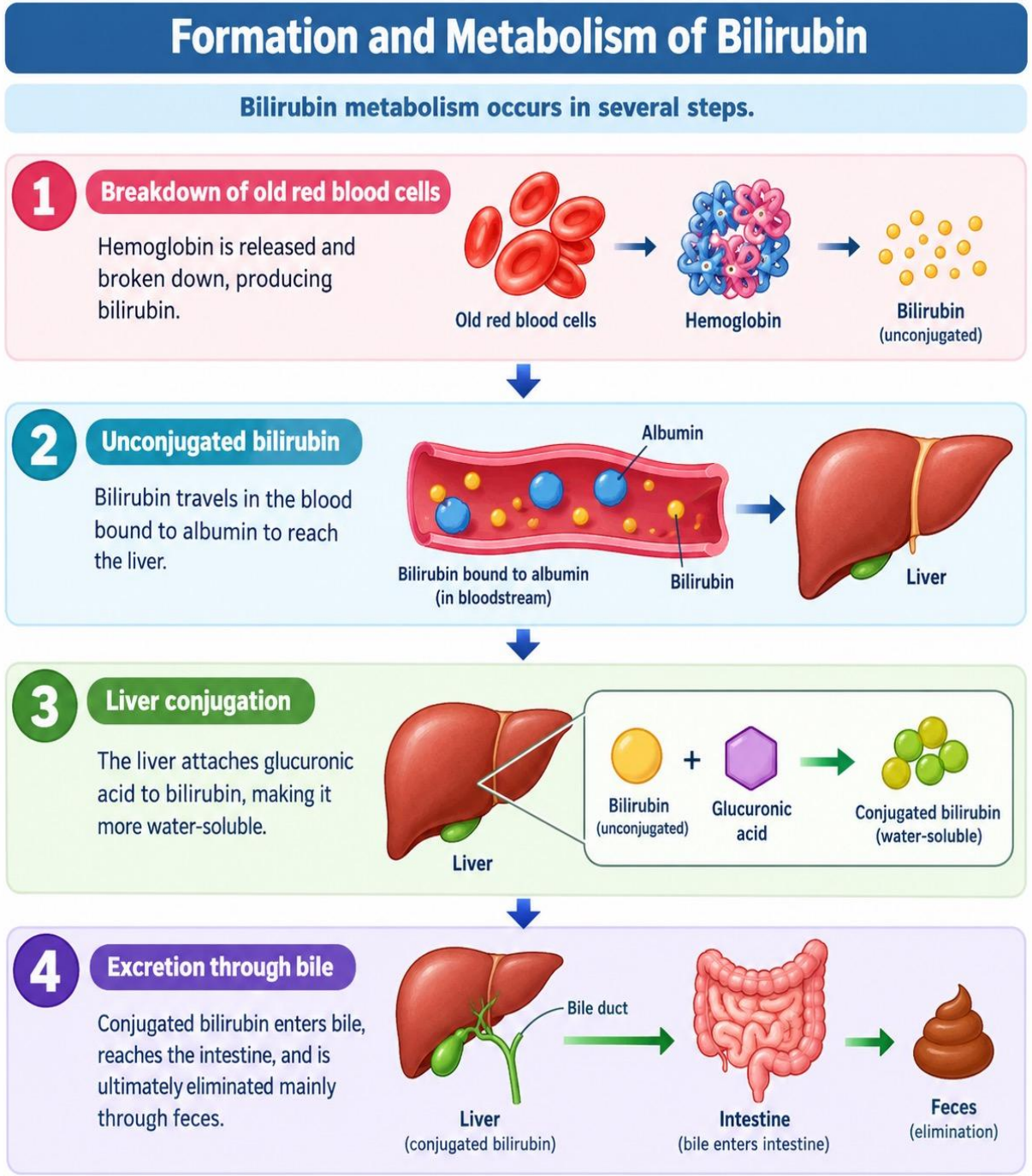




Bilirubin

- Bilirubin is a yellow pigment produced mainly when old red blood cells are broken down. The liver processes bilirubin and helps remove it from the body through bile.





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- **Types of bilirubin**

1. Unconjugated bilirubin (indirect bilirubin)

- - Produced before processing by the liver.
- - Not water-soluble.
- - Travels in the blood bound to albumin.
- - Cannot normally be excreted directly into urine.

2. Conjugated bilirubin (direct bilirubin)

- - Formed in the liver when bilirubin is conjugated with glucuronic acid.
- - More water-soluble.
- - Normally secreted into bile.
- - May appear in urine when blood levels rise due to certain liver or bile duct disorders.

3. Total bilirubin

Total bilirubin is the sum of conjugated and unconjugated bilirubin measured in the blood.

Causes of increased bilirubin

Type	**Possible causes**
Increased unconjugated bilirubin	Excessive red blood cell break-down, Gilbert syndrome, or impaired bilirubi n conjugation
Increased conjugated bilirubin	Bile duct obstruction,cholestasis, or certain liver diseases
Increased total bilirubin	May occur in jaundice due to several different causes

Bilirubin Fractionation

Bilirubin fractionation is a blood test that measures bilirubin and differentiates it into **unconjugated** and **conjugated** fractions. It helps assess liver function, hemolysis and problems with bile flow.

Unconjugated Bilirubin (Indirect)	Feature	Conjugated Bilirubin (Direct)
Produced from heme breakdown, mainly in macrophages (reticuloendothelial system).	Formation	Formed in the liver by conjugation with glucuronic acid.
Not water-soluble (lipophilic).	Solubility	**Water-soluble** (hydrophilic).
Bound to albumin in blood.	Albumin binding	Usually not tightly bound to albumin.
Not yet processed by the liver.	Liver processing	Conjugated with **glucuronic acid** (in the liver).
Not normally excreted in urine.	Excretion in urine	Can be excreted in **urine** if elevated in blood.
Hemolysis or impaired conjugation (e.g., Gilbert syndrome).	Increased levels may suggest	Liver disease or bile duct obstruction (cholestasis).

Normal Serum Bilirubin Levels
(in adults)

Bilirubin Fraction	Normal Reference Level
Total bilirubin	0.1 – 1.2 mg/dL
Conjugated (direct) bilirubin	< 0.3 mg/dL
Unconjugated (indirect) bilirubin	0.2 – 0.8 mg/dL

Note: Ranges may vary slightly between laboratories.

Formula for Bilirubin Fractionation

Indirect bilirubin = Total bilirubin – Direct bilirubin

Example:

Total bilirubin = 1.0 mg/dL
Direct bilirubin = 0.2 mg/dL

Indirect bilirubin = 1.0 – 0.2
= 0.8 mg/dL

Key Points to Remember

- **Unconjugated bilirubin** is water-insoluble and binds to albumin.
- **Conjugated bilirubin** is water-soluble and can appear in urine when elevated.
- **Increased unconjugated bilirubin:** hemolysis or reduced conjugation (e.g., Gilbert syndrome).
- **Increased conjugated bilirubin:** liver disease or bile duct obstruction.

Bilirubin fractionation helps identify the underlying cause of jaundice.

Jaundice

Jaundice is the yellow discoloration of the skin & whites of the eyes caused by increased bilirubin in the blood.

Clinical significance:

Bilirubin testing helps investigate jaundice, red blood cell breakdown, liver disorders, and obstruction of bile flow.

JAUNDICE

Jaundice is the yellow discoloration of the skin, sclera (whites of the eyes), and mucous membranes due to elevated levels of bilirubin in the blood.

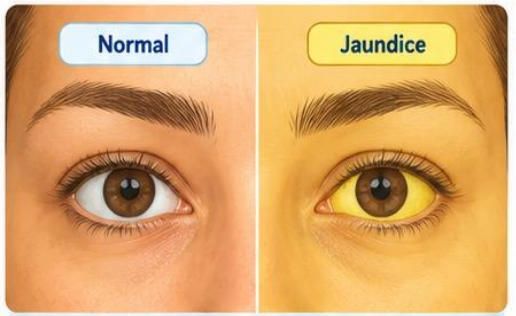
Remember:
- Bilirubin gives the yellow color.
- Jaundice is a symptom, not a disease.

Causes

1. Increased production (Hemolysis)
- Hemolytic anemia
- Blood transfusion reactions
- Malaria

2. Impaired liver processing (Hepatocellular)
- Viral hepatitis (A, B, C, etc.)
- Alcoholic liver disease
- Fatty liver disease
- Drug-induced liver injury

3. Obstruction of bile flow (Cholestatic)
- Gallstones
- Bile duct stricture
- Pancreatic or biliary tumors
- Primary biliary cholangitis



Yellow discoloration of skin and sclera in jaundice

Common Symptoms
- Yellowing of skin, eyes and mucous membranes
- Dark urine (due to conjugated bilirubin)
- Pale or clay-colored stools (lack of stercobilin)
- Itching (in cholestasis)
- Fatigue, loss of appetite (in liver disease)

Types of Jaundice

1. Pre-hepatic (Hemolytic)
- Excess bilirubin production
- ↑ Unconjugated bilirubin
- Normal liver enzymes
- Dark urine, normal-colored stools

2. Hepatic (Hepatocellular)
- Impaired liver uptake, conjugation or excretion
- ↑ Both unconjugated and conjugated bilirubin
- ↑ Liver enzymes (ALT, AST)
- Dark urine, pale stools

3. Post-hepatic (Obstructive)
- Blocked bile flow
- ↑ Conjugated bilirubin
- ↑ ALP and GGT
- Dark urine, pale/clay-colored stools
- Itching (common)

Diagnostic Approach

1. History & Examination
- Onset, duration
- Risk factors (alcohol, drugs, viral exposure)
- Associated symptoms

2. Laboratory Tests
- Bilirubin (total, direct, indirect)
- Liver function tests (ALT, AST, ALP, GGT)
- CBC, PT/INR
- Urine (bilirubin, urobilinogen)

3. Imaging (if needed)
- Ultrasound (liver, gallbadder, bile ducts)
- CT/MRI/ERCP (for obstruction)

Interpretation of Urine and Stool

Urine
- Conjugated bilirubin present → dark urine
- Urobilinogen ↑ (in hemolysis or liver disease)
- May be absent in complete bile duct obstruction

Stool
- Normal → brown color
- Pale/clay-colored → suggests obstruction (lack of stercobilin)

Key Points to Remember
- Jaundice is a sign of elevated bilirubin in the blood.
- It helps identify underlying cause (hemolysis, liver disease, or obstruction).
- Conjugated bilirubin → water-soluble (can appear in urine).
- Unconjugated bilirubin → not water-soluble (does not appear in urine).
- Early evaluation and treatment prevent complications.

Differential Interpretation of Jaundice

Jaundice Type	Conjugated (Direct)	Unconjugated (Indirect)	SGPT (ALT)	ALP	GGT
Pre-Hepatic	Normal	**Elevated (↑)**	Normal	Normal	Normal
Hepatic	**Elevated (↑)**	**Elevated (↑)**	**Marked (↑↑)**	Elevated / Normal	Normal
Post-Hepatic	**Marked (↑↑↑)**	**Normal**	Modest Increase	**Marked (↑↑)**	**Severe (↑↑↑)**

Interpretation of LFT results

Standard Patient Panel

Test Analyte	Observed Result	Reference Range
Total Bilirubin	0.9 mg/dL	0.2 – 1.2 mg/dL
Direct Bilirubin	0.2 mg/dL	0.0 – 0.3 mg/dL
Indirect Bilirubin	0.7 mg/dL	0.25 – 0.9 mg/dL
SGPT (ALT)	19 U/L	< 45 U/L
Alkaline Phosphatase	60 U/L	53 – 128 U/L
Gamma GT	10 U/L	< 55 U/L
SGOT (AST)	26 U/L	< 35 U/L

Diagnostic Finding

Normal LFT Profile

All liver enzymes, synthetic markers, and bilirubin fractions lie strictly within physiological reference limits.

Clinical Interpretation Summary

Hepatocellular Patterns

Marked ALT & AST elevation highlights primary liver parenchyma injury.

Cholestatic Patterns

Concomitant ALP & GGT elevation indicates biliary tree obstruction.

Urine Bilirubin Significance

Bilirubinuria strictly signals elevated conjugated direct serum bilirubin.

THANK YOU

